



Quantum cascade laser imaging (LDIR) and machine learning for the identification of environmentally exposed microplastics and polymers

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ABSTRACT

Monitoring of microplastics in environmental samples is relevant to the scientific world, as well as to environmental agencies and water authorities, in particular considering increasing efforts to decrease emissions and the growing concern of governments and the public. Therefore, rapid accurate detection and identification of microplastics including polymers, despite their degradation in the environment, is crucial. The degradation has a significant impact on the infrared spectra of the microplastics and can impede the identification process. This work presents a novel approach to addressing the problem of identification of weathered microplastics. A quantum cascade laser (LDIR) was used to record the infrared spectra of various polymeric particles (81,291 individual particles). Using a combination of pristine and weathered particles, two supervised machine learning (ML) models, namely Subspace k-Nearest Neighbor (Sub-kNN) and Boosted Decision Tree (BDT), were trained to recognize the spectrum characteristics of labeled particles and then used to identify unlabeled samples, with an identification accuracy of 89.7% and 77.1% using 10-fold cross validation. About 90% of the samples could be identified via the Sub-kNN or BDT models. Subsequently, a non-supervised ML model, namely, Density-based Spatial Clustering of Applications with Noise (DBSCAN), was used to cluster samples which could not be labeled from the supervised ML model. This enabled the detection of additional subgroups of microplastics. Manual labelling can then be carried out on a selection of spectra per group (e.g., centroids of each cluster), hence accelerating the identification process and allowing to add new labeled samples to the initial supervised ML. Although expert efforts are still needed, the proposed method greatly lowers labeling efforts by using the combined supervised and unsupervised learning models. In the future, the use of deep neural networks could further boost the implementation of these kinds of approaches for polymer and microplastic identification in environmental settings.

1. Introduction

Polymers, which include microplastics, have been detected in oceans, rivers, lakes, bottled water and to a certain extent also in drinking water (Mintenig et al., 2020; Oßmann et al., 2018; Xu et al., 2019; Lorenz et al., 2019; Oßmann, 2021). Research about the occurrence of microplastics in water, especially drinking water, is highly relevant given that they are explicitly mentioned in regulations such as the new European Drinking Water Directive (DWD), which was adopted in December 2020 and came into force on January 12, 2021 (Schymanski et al., 2021; The European parliament and the, 2020). The latter mandates that Member States must implement the provisions of the revised DWD into national laws and regulations by January 12, 2024. The directive aims to address growing public concern about the effects

of microplastics on human health, in particular if exposure occurs through ingestion of water intended for human consumption and to address the risks involved. This means that microplastics must be monitored. Risk assessments can then be carried out on the basis of the data. Fast and reliable analysis methods and knowledge about the occurrence of microplastics are hence crucial.

Several optical techniques exist to detect and characterize polymers and microplastics, such as Fourier Transform Infrared (FTIR), Raman and Laser Direct Infrared (LDIR) spectroscopy (Silva et al., 2018; Primpke et al., 2021). These techniques rely on databases containing reference spectra to which acquired particle spectra are compared. Reference spectra are generally collected from pristine polymers, however, particles detected in environmental samples are seldom pristine and have often undergone more or less extensive weathering due to

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environmental conditions (e.g., UV-light, oxidation). Weathering will affect their physicochemical properties, and hence will influence their interaction with infrared light. Ultimately, this will translate in modified spectral profiles, which can potentially lead to misidentifications (Liu et al., 2020). Compiling a database with spectra of polymers with different degrees of weathering would be extremely time-consuming and is not a viable option. Therefore, alternative approaches combining more rapid analytical techniques, as well as more advanced data analysis approaches, which take advantage of existing data from both pristine and weathered samples, are necessary.

LDIR has recently been introduced as an alternative technique for the analysis of polymers and microplastics in environmental samples. Compared to approaches such as FTIR, it has the advantage of scanning samples prior to actual imaging and eventually only analyses areas with actual particles (Dong et al., 2021). This has obvious advantages in terms of analysis time, as spectra will be acquired when particles have been detected during the preliminary scan. Samples that are almost void of particles are therefore measured considerably faster with LDIR. One downside of this is also that if two particles are adjacent to each other, they can be seen as one particle as per particle only one spectrum is recorded. μ FTIR has a better chance to detect that there are two individual particles as the whole particle area is analyzed. If the two particles have distinctively different spectra, the μ FTIR software will count them as two particles. Another disadvantage of the LDIR is that the infrared band recorded with LDIR instruments is narrower (ca. 1800 to 900 cm^{-1}) (Primpke et al., 2020). As less information is collected, LDIR is more prone to misidentification when analyzing weathered particles compared to μ FTIR (Primpke et al., 2020). Therefore, more effective classification methods are required to minimize the risk of misidentification. Machine learning (ML) can help improve classification, yet its implementation in the field of microplastics identification is still limited. ML refers to a collection of methods derived from statistics and computer science with the goal of training mathematical models to make predictions or decisions based on observed data (Ng, 2018). ML, particularly its sub-branch deep learning and reinforcement learning, has greatly gained in popularity, becoming the backbone of various scientific fields and industrial sectors, including biomedical engineering and water research (Mowbray et al., 2021; Sha et al., 2021; Shen et al., 2018).

When the output of a dataset has been manually labeled prior to training, supervised ML is often used to determine which inputs are the major factors (also known as features) that contribute to the labeled output. When no labels are available, unsupervised learning is utilized to cluster samples. In practice, one often encounters a mixed dataset with a small number of labeled samples and a large number of unlabeled samples, prompting the use of active learning (Géron, 2019). Active learning first seeks to develop a supervised model using labeled data and then apply it to classify unlabeled data. Next, the samples for which the model prediction is most uncertain are sorted out and to be handled by experts. The labelling effort iterates until the model achieves a desired level of model performance. In doing so, it is possible to use a relatively small number of labeled samples to find the type of polymers from the unlabeled samples.

The goal of this work was to develop an effective approach to identify environmentally exposed polymers using a combination of LDIR and ML. Available databases of pristine polymers and other plastics were collated to a large dataset (81 k samples) of spectra of environmentally exposed polymers and microplastics collected from surface and drinking water across the Netherlands. Using the available labeled examples as input, a supervised ML model was trained to build a classifier to identify polymer types based on their spectra, which was then applied to unlabeled samples. The latter was accepted and labeled with their appropriate types, only when the classifier determined their types with a probability larger than 70%. Finally, an unsupervised ML algorithm was used to cluster unaccepted samples and domain experts could determine the type of cluster by examining a small number of representative

samples from each cluster.

2. Materials and methods

2.1. Experiments

2.1.1. Experiment chemicals

The following chemicals were used in this study: sodium dodecylsulfonate and KOH were purchased from Merck (Darmstadt, Germany); H_2O_2 (30%), ZnCl_2 and Ethanol were purchased from Boom (Meppel, the Netherlands) and fluorescent green polyethylene microspheres were purchased from Cospheric (Santa Barbara, USA). Ultrapure water was made using an ELGA system (18.2 $\text{M}\Omega/\text{cm}$, LabWater, Lane End, UK). All solvents used, including the cleaning solvents, were filtered prior to usage over 5 μm sieves and stored in closed bottles. All solvents were exclusively used for polymer analysis to minimize the risk of cross-contamination. All equipment was at all times covered when not used to prevent air contamination, except when the samples were taken on location.

2.1.2. Particle analysis

Samples were analyzed using the quantum cascade laser (QCL) based Agilent chemical imaging laser direct infrared (LDIR) system. Wavenumbers from 975 to 1800 cm^{-1} with a resolution of 0.5 cm^{-1} are covered. Particles ranging from 20 to 500 μm were measured and counted. The smallest particles which can be analyzed with the available instrument would possibly be 3 μm or so.

2.1.3. Sampling

Two sampling methods were used here, namely a cascade of metal sieves and a metal candle filter cascade. The first one was used for samples with an expected high number of particles, the latter for samples with an expected low number of particles (additional information is provided in the Supporting Information). Samples were taken from different surface waters (river Rhine, river Meus, river Overijsselse Vecht), groundwater and at different stages of drinking water treatment. At each sampling event, between 1000 and 10,000 L of water were filtered through either the cascade of two metal sieves (Gilson, USA) of 500 μm and 100 μm , with a 10 μm plankton net (Hydro-Bios, Germany) at the end or the candle filter cascade (5, 10, 100 and 500 μm). The 500 μm sieve or filter were used to remove particles and prevent clogging of the smaller sieves/filters. Particles of 500 μm or larger were not analyzed. Residues from other sieves/filters were transferred into separate glass bottles using Milli-Q ultrapure water (Millipore Sigma, USA) and stored at 4 $^{\circ}\text{C}$.

2.1.4. Quality assurance

Potentially, contamination occurring during sample handling was minimized by cleaning all laboratory surfaces with ethanol. Equipment was rinsed with ethanol and covered immediately with aluminium foil, and a cotton laboratory coat was worn at all times by the analysts involved in sample preparation and analysis. In addition, all solvents and solutions were filtered prior to use (10 μm stainless steel mesh). Used materials were not made from plastic wherever possible (e.g., a metal filter setup with a Teflon tube, the separation funnels made of glass). Negative controls were treated in parallel to each batch of actual samples to determine the degree of contamination. For the positive control, a known number of plastic particles (green fluorescent PE, average diameter 100 μm) were added to a water sample. The percentage of particle number before and after work-up was used to estimate of the recovery rates. These control particles were counted visually under a microscope.

2.1.5. Sample processing and analysis

Particle analyses were based on previously described methods (Minteni et al., 2020). Particle analysis focused on 10 μm and 100 μm

residues of the sieves. The suspensions from these two fractions were combined and filtered over a 10 μm metal mesh. The latter was then transferred into a beaker with a 10% sodium dodecylsulfonate solution. After 24 h, the suspension was filtered again over a 10 μm metal mesh and transferred into a beaker with 75 ml 12.5% potassium hydroxide solution and left standing for 5 days at 35 °C. Subsequently, the suspension was filtered again through the 10 μm metal filter. The residues were then transferred into a beaker with 50 ml 30% hydrogen peroxide solution and left standing for 24 h at 35 °C. The sample was filtered again through the same metal filter, and the residues were transferred into a separation funnel using a 100 ml zinc chloride solution (1.6 g/cm³). The funnels were shaken and left standing to enable settling of denser materials. The settled material was discarded by continuously turning the valve of the funnel to prevent clogging, re-suspension and loss of plastics. About 10 mL liquid was allowed to remain in the funnel. These 10 mL were filtered again over a 10 μm metal filter. Using four mL ethanol, the retained materials were removed from the filter and transferred into a glass vial. A vortex was created in this suspension to distribute the particles evenly. An aliquot (2 × 50 μL) was taken and transferred onto a microscope plate for analysis. Aliquots were used to avoid having too high a particle density on the plates. Total number of particles per sample were extrapolated from these aliquots. Samples were analyzed using the LDIR. Between four to six samples were analyzed in parallel in combination with a negative control sample to detect (cross)-contamination during work-up and a sample spiked with a known amount of clearly visible fluorescent polyethylene microbeads to calculate the recovery rate (positive control). After each set, the whole equipment was cleaned using ultrapure water and ethanol. The equipment was then covered with aluminium foil to prevent contamination from the air.

2.1.6. Types of particles

In total, 81,291 particles from various Dutch aqueous matrices (drinking water, surface water, effluent from wastewater treatment plants) were used in this study, of which 223 were either pristine polymers or had already been manually labeled ($\approx 0.3\%$) and 81,068 being unlabeled ($\approx 99.7\%$). Ten different types of polymers are among the labeled particles, including cellulosic, polyacetal (POM), polyamide (PA), polyethylene (PE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyurethane (PU), polyvinylchloride (PVC), and silica. An additional category called non-plastic particle was also added. Other types of polymers were not the main focus of the study and were regarded as outliers to these 11 predefined categories when making predictions based on a classification model. These polymers were chosen as they are the most commonly found in the environment according to recent literature (Mintenig et al., 2020; Scherer et al., 2020). In the future more polymers will be added.

2.2. Machine learning models

Prior to introducing the ML models, it is important to specify the input and output. In this study, 1,650 features are used as input, which corresponds to absorption rates on wave numbers ranging from 975 to 1800 cm^{-1} with a resolution of 0.5 cm^{-1} , where Supplementary Material I (Figure S11) shows an example of spectroscopes of polymers. The model output stands for the polymer type. In other words, the ML model aims to predict the polymer type based on 1,650 features. Moreover, due to the uneven distribution of labeled and unlabeled samples, we combined several supervised and unsupervised models to predict polymer types (Table 1). Note we use the words labeled and classified to represent polymer types that are manually annotated and determined by the algorithms, respectively. Fig. 1 shows the ML models adopted in this study, which comprises of the following components:

- Data preprocessing: We pre-processed the data for better training the models. First, samples whose maximum absorption value across all

Table 1

List of main models used in this study.

	Name	Type	Brief introduction and functions
1	subspace k-nearest neighbor	ensemble supervised learning	The random subspace k-nearest neighbor method is an ensemble ML method, which often achieves high accuracy when used to solve classification problems. We build the subspace-KNN model in Matlab using the function <i>fitcensemble</i> .
2	boosted decision trees	ensemble supervised learning	By fitting the residuals of preceding decision trees, the boosting algorithm iterates to learn from data. We built the boosted decision tree model in Matlab using the function <i>fitcensemble</i> .
3	density-based spatial clustering of applications with noise	unsupervised learning	DBSCAN is a non-parametric clustering algorithm based on calculations of densities. We build the DBSCAN model in Matlab using the function <i>dbscan</i> .
4	t-distributed stochastic neighbor embedding	unsupervised learning	The t-sne algorithm is an ML technique for reducing the dimensionality of data space. We build the subspace-KNN model in Matlab using the function <i>tsne</i> .

wave numbers was smaller than 0.01 cm^{-1} were removed as they likely involved measurement of non-polymeric particles (e.g., air bubbles). Next, the values were standardized to their mean value and unit variance for each wave number. In other words, normalized features were used to train the models.

- Training a classification model: Specifically, a supervised model (ensemble learning) was developed, aiming to classify 223 labeled samples into 11 pre-defined polymer types. Two commonly used classification models were trained and compared, including (random) subspace k-nearest neighbor (Peterson and L. K-Nearest Neighbor, 2009), and boosted decision trees (Breiman et al., 2017), i. e., Model #1 and #2 in Table 1. To achieve the optimal model performance, their hyperparameters were tuned by implementing a grid search for (i) the learning rates (i.e., the step that an optimization algorithm takes to minimize its loss function) and (ii) the number of weak learners (i.e., a predictor that performs relatively poor and is aggregated with other predictors to generate a well-performing predictor). To evaluate the performance of the supervised models, the classification accuracy (= 1 - re-substitution error) is used as the performance indicator. The accuracy of a classifier is defined as the ratio of correctly identified samples to the total number of samples. Besides, to prevent overfitting the model, 10-fold cross validation was also performed, which was conducted by randomly partitioning the training dataset into 10 equal-sized subsets and computing the mean value of the accuracies of all subsets. In other words, total performance was measured as the accuracy of the cross-validated model based on ten subsets. Note that cross validation provides a rough estimate of the accuracy when applying the trained ML model to new data.
- Application of the classification model: The trained classification model was applied to all unlabeled samples. Particularly, the classification model computed the probability that a sample fell into a given category (i.e., polymer type). As a result, a minimum acceptance criterion (e.g., 70% or 85%) was specified, over which samples were considered classifiable and below which they were considered (model) outliers not belonging to any of the 11 pre-defined classes. In

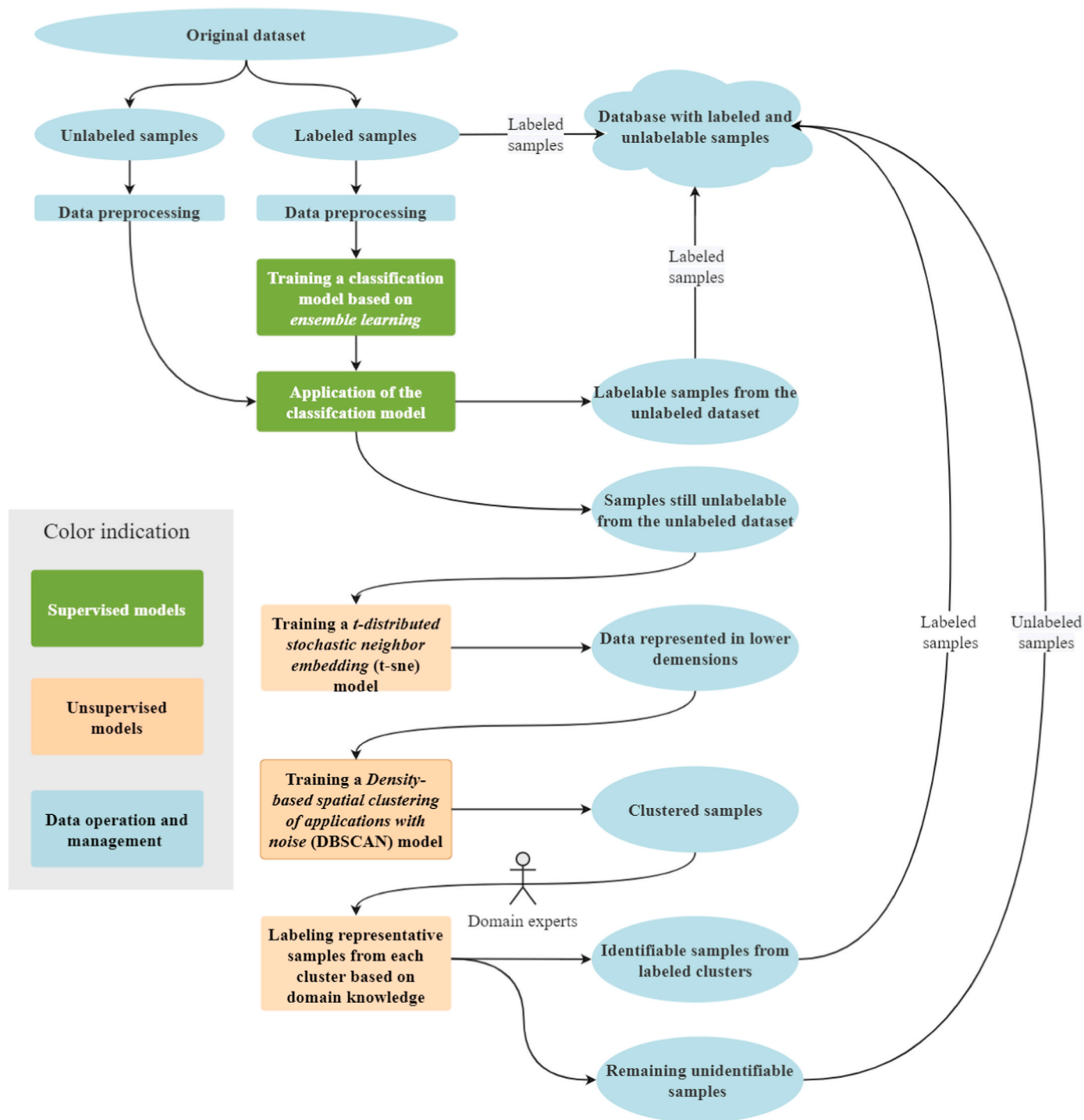


Fig. 1. The framework to train combined ML models. In the diagram, the shapes of oval, rectangular, and cloud indicate input/output, processes/models, and database respectively.

other words, outliers are those samples whose classification model cannot detect their types.

- Training a clustering model: Outlier samples are still unlabeled and require additional analysis using unsupervised learning, for which the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) model was used (Schubert et al., 2017). The DBSCAN algorithm (model #3 shown in Table 1) is an efficient method for grouping data with similar features and meanwhile recognizing noises, i.e., independent sample points not belonging to any cluster.

In doing so, samples with similar spectra patterns can be identified in the same cluster.

- Labelling the clusters: Finally, the central points of each cluster can be manually assessed by an expert operator to determine the type of polymer. Samples within the same cluster were then assumed to be of the same type and are labeled as such. As a result, unlabeled samples were split into two sets: identifiable samples from clusters and unidentifiable samples as outliers.
- Additionally, since the number of features is 1,650, we adopted an unsupervised learning approach, namely t-distributed Stochastic

Neighbor Embedding (t-sne, model #4 shown in Table 1), to visualize the distance between samples in a 2D plane (Belkina et al., 2019). Note that when implementing t-sne, the Kullback-Leibler divergence, which described the loss from input to output, was kept below 25%.

3. Results and discussion

3.1. Classification of labeled polymer samples

Two classification models (Model 1 and Model 2 shown in Table 1) were trained on 223 labeled samples spanning 11 different polymer types. In general, the subspace KNN model outperformed the boosted decision tree model, with the former achieving 89.7% accuracy (12.6% higher than that of the latter). Fig. 2 shows the confusion matrix for the two models, namely, a subspace KNN model and a boosted decision tree model. Rows represent true classes (predefined labels) whereas columns represent predicted classes (labels marked by the classification models). The subspace KNN model can predict the class of all polymer samples with no more than 3 mislabeled samples for each type. The best prediction was obtained for PE, as all samples were correctly labeled. The worst prediction was obtained for POM, where 2 out of 5 samples were mislabeled. Besides, 5 out of 15 non-plastic samples were mislabeled ($\approx 33.3\%$). This high inaccuracy is due to the fact that non-plastic particles do not have a distinct spectrum and only a limited number of spectra were available for training the classifiers. On the other hand, the boosted decision tree performed worse than the subspace KNN model, in particular because 22 particles were mislabeled as silica. This was because the boosted decision tree algorithm underfits the decision region for silica. Apart from the mislabeled silica, the boosted decision tree performs generally well, though it is slightly worse than the subspace KNN model across all polymer types. Based on these results, the subspace KNN model was deemed the most appropriate for classifying polymers in our case study.

3.2. Classification of unlabeled samples

The previously trained classification model was then applied to unlabeled samples (real samples). The model produced 11 probabilities describing how likely an unknown polymer belongs to one of the 11 polymer types. Therefore, a threshold to accept the prediction probability had to be defined. A range of possible threshold values ranging

from 67.5% to 100% was evaluated and the corresponding number of unaccepted samples was computed. The greater the threshold value, the larger the number of unaccepted predicted labels, as can be seen in Fig. 3. For the purpose of the applications considered in this study, 10% of new samples were deemed by the domain expert as acceptable. Hence, in Fig. 3, a threshold of 0.7 was selected. Obviously, this threshold was defined subjectively, and different values should be selected depending on what is considered acceptable for the specific application. Furthermore, if more information about the occurrence of different types of polymers in samples becomes available, the threshold can be adjusted accordingly. Supplementary Material II shows a supplemental analysis to help determine the threshold value (Figure SI2).

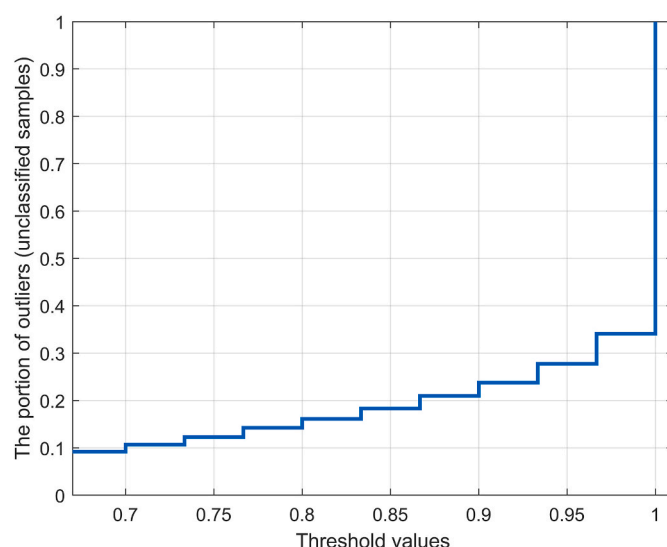


Fig. 3. Percentage of unclassifiable samples as a function of the threshold for accepting the prediction of polymer types. The higher the threshold value is, the more samples are not accepted as labeled samples. This study uses the threshold of 0.7, leading to approximately 10% of unclassifiable samples (c.a. 8,100 samples).

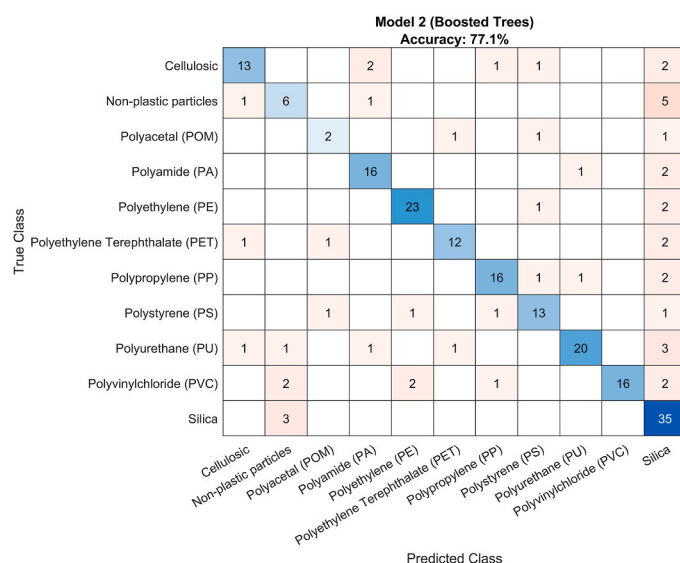
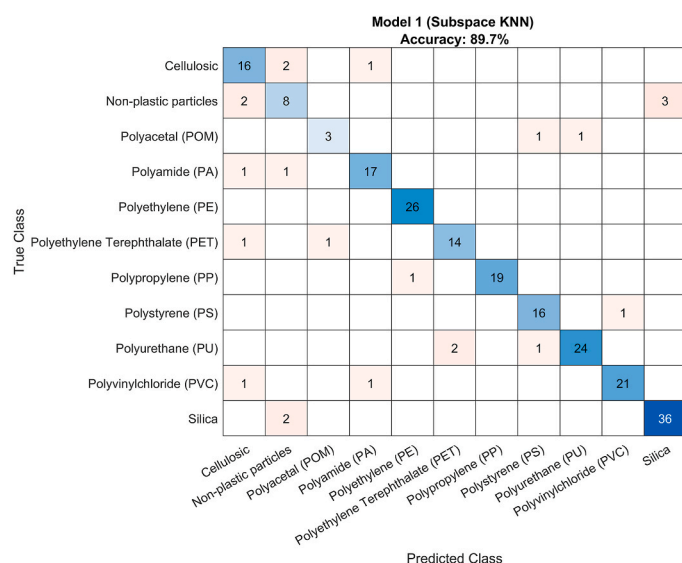


Fig. 2. Results of classification model with 10-fold cross validation. Two models, subspace KNN and boosted decision tree, were developed and compared. Subspace KNN (accuracy = 89.7%) outperforms boosted decision tree (accuracy = 77.1%) for our classification problem.

3.3. Clustering unlabeled samples

The third part of the developed approach focused on the remaining 10% of particles which could not be classified using the 0.7 probability and required extra processing. As described in Fig. 1, the unsupervised model DBSCAN (model #3 shown in Table 1) was used to cluster samples. Two parameters need to be set before training the model, namely the minimum number of points (denoted as minpts) and the neighborhood searching radius (denoted as eps). Different combinations of these two parameters can lead to the number of clustered types and the number of outliers (samples not belonging to any cluster). When the values for eps and minpts are fixed, one or more central points can be determined. These points represent those that contain at least minpts points inside the radius eps. It is, however, a pragmatic choice to determine these two values. Fig. 4 illustrates three possible combinations of eps and minpts values that provide different clustering results. When the value of eps is increased, the number of outliers decreases as more samples can be clustered within a broader searching radius. When the value of minpts is decreased, more clusters and fewer outliers appear, as more clusters of lower sizes can emerge. However, it is not a straightforward linear relationship. Supplementary Material III (Figure S13) shows a diagram that can be used to help determine the values subjectively. Assuming that one should have 30 polymer types, with at least 30 samples per type, and roughly 40% outliers that are minor types and cannot be clustered, one can find a compromise using the mentioned diagram (in this instance, $\text{eps} = 2.75$ and $\text{minpts} = 30$). By clustering these data, domain experts need to label a smaller number

of samples with far less work. Due to the fact that this is a manual check performed by a domain expert, in which the domain expert must screen a large number of spectra and determine the type of polymers, we do not demonstrate the manual process in this study.

3.4. The implications of machine learning models

3.4.1. Selection of ML models

In Section 3.1, two ML models, namely, subspace KNN and boosted decision tree, were trained to classify 223 samples into 11 pre-defined categories, whose accuracy of predictions were 89.7% and 77.1% respectively. Besides these two models, we also trained other ensemble ML models for comparison, including bagged decision tree (66.1%), ensemble subspace discriminant (61%), weighted KNN (67%), and RUSBoost decision tree (8%), which all performed worse than the two reported models. As a result, we only showed detailed results in Section 3.1 and considered the subspace KNN model as the best classifier for our polymer samples. Typically, it is difficult to know which model has the best performance before training and comparing candidate models, as the no free lunch theorem implies (Wolpert and Macready, 1997). We suggest that the above-mentioned ensemble ML models be trained and compared before deploying an ML model for a new application.

Besides the ML models that are used in this study, deep neural networks can be utilized to classify polymers. For example, an autoencoder that can encode and decode the high-dimensional feature space and then predict polymer types may be applicable in our case. However, training such a model requires a significant number of samples, which were

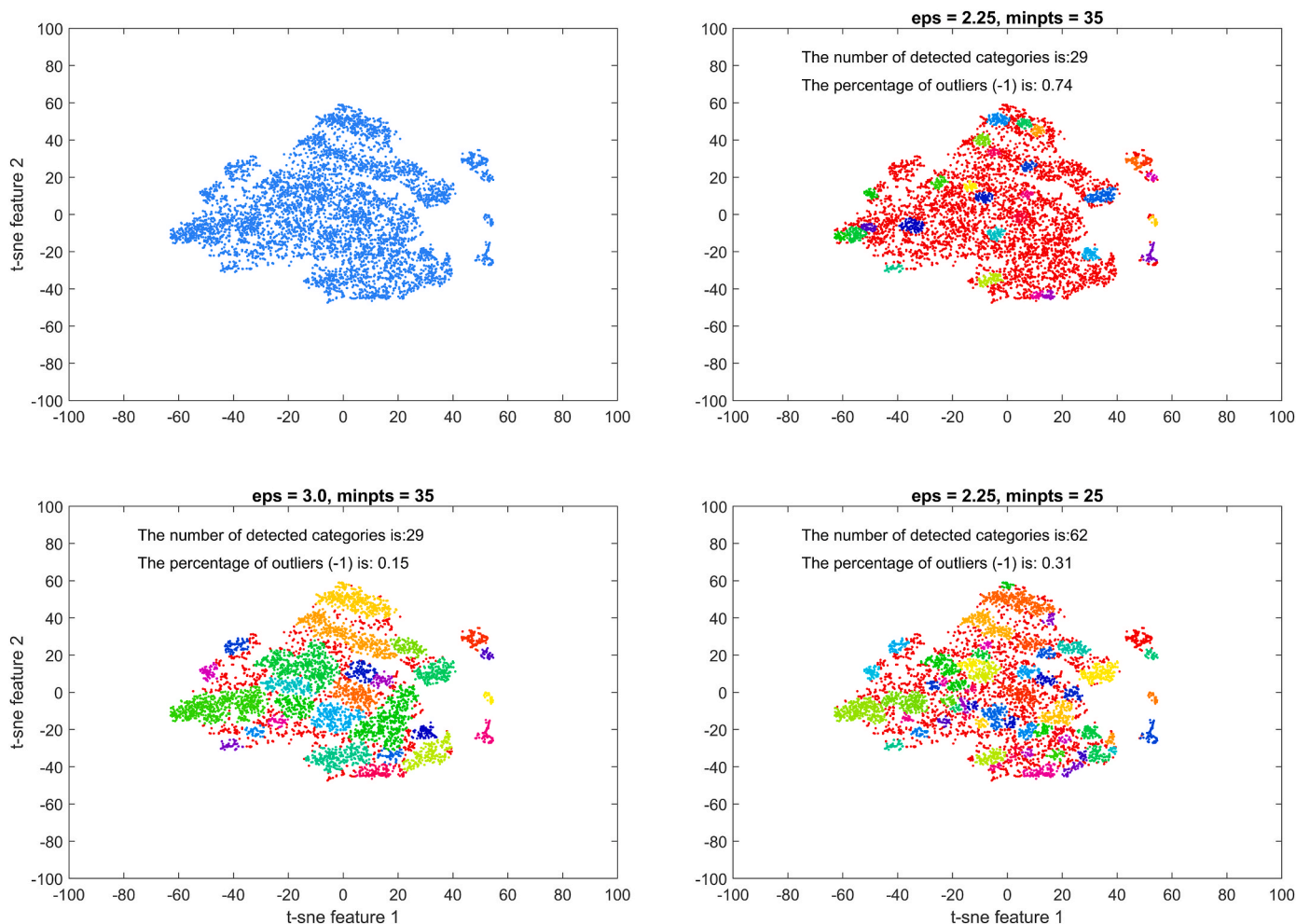


Fig. 4. Clustering unlabeled samples based on two parameters (the distance and the minimum number of points contained in one cluster).

unavailable at the time of conducting this research. In future study, we intend to collect more samples and employ deep neural networks to address this issue.

3.4.2. Selection of features

In this study, absorption rates on wave numbers ranging from 975 to 1,800 cm^{-1} were used as features to train the classification model. Alternatively, the 1st-order derivative of the absorption rate is often used, too. To compare with the original absorption rate, a subspace KNN model and a boosted decision tree model were trained based on the 1st-order derivative of the absorption rate, resulting in the prediction accuracy of 92% and 17% respectively. The boosted decision model is much worse than the one trained on the original absorption rate while the subspace KNN is slightly better. As a result, we did not adopt the 1st-order derivative for the classification and clustering analyses because (i) the performance only improves insignificantly for the subspace KNN model; (ii) to apply the classification model to new samples, it takes extra computation time to calculate the 1st-order derivative; (iii) it is more intuitive to interpret the original absorption rate than the 1st order derivative.

3.4.3. The size of dataset for training an ML model

ML is a data-driven technique requiring a large number of observations, particularly in the training dataset, in order to develop a good model. In this work, initially we started with a dataset of 60 labeled samples and only achieved an accuracy of approximately 70%, which is 19.7% lower than the model trained on the 223 labeled samples. Ideally, adding extra samples would improve the model performance. There is, however, the concern of finding a balance between model performance and manual labelling effort. In practice, one can iterate the process of adding samples and analyze the model's performance as it evolves. When performance exceeds a predetermined level or when adding new samples improves performance little, no additional samples are required. Additionally, if a large number of samples are misclassified for a particular class, extra labeled samples for that class should be added.

Consider the case of non-plastic polymers encountered in this study: it remains challenging for the classification model to accurately classify non-plastic samples type because (1) there is an infinite number of different non-plastic matter, (2) especially inorganic non-plastic samples, unlike plastic samples, can lack substantial group characteristics and (3) certain non-plastic samples may present characteristics that are very similar to those of other groups and hence can be falsely labeled as plastic. As a result, the model developer and user should consider the accuracy of categorizing a non-plastic polymer when applying the classification model or exclude non-plastic samples entirely and regard them as outliers.

3.4.4. Implementation of polymer screening

To conduct the polymers screening, one should first train a classification model, then detect unclassifiable samples (outliers), and finally build a clustering model to group outliers. Note that 90% of the unlabeled samples ($\approx 73,100$) in this study can be successfully identified based on the 223 labeled examples. Thus, considerable effort for the domain expert to label samples can be saved. The remaining 10% of the outlier samples ($\approx 8,100$) can be clustered and subsequently labeled by domain experts. The amount of effort required to identify polymer types is dependent on the number of clusters and representative samples within each cluster. One who aims to adopt this method must weigh the trade-off between the overall performance and the time required to process individual samples. If the classifiable samples (≈ 90 percent) already match the criteria for conducting polymer screening, training a clustering model and manual labelling can also be omitted.

4. Conclusion

In this study, we investigated the applicability of infrared data from

an LDIR in combination with ML to identify polymers in water samples. This novel approach is necessary as the number of plastics at various stages of degradation in the environment and other aqueous samples will be unlimited. A database with a certain amount of reference spectra alone and comparison using Pearson correlation coefficient will not suffice. The major finding was that ML, namely Subspace k-Nearest Neighbor (Sub-kNN) and Boosted Decision Tree (BDT), was able to further identify the type of polymers, based on the absorption measured by LDIR with an identification accuracy of 89.7% and 77.1% using 10-fold cross validation. This research shows a suitable way to identify polymers that are not pristine and have been weathered. Creating realistically weathered particles (each type of polymer at different stages of weathering) is not a viable option. For our research we used a limited number of weathered particles from the environment to train a supervised ML model. This model builds a classifier to identify polymer types based on their spectra, which was then applied to unlabeled samples. These findings suggest that in general artificial-intelligence-aided techniques can greatly support and promote analytical chemistry and water research, which is of interest to researchers who aim to classify polymers (based on classification models) or predict their bio-chemical behaviors (based on regression models). Nevertheless, the application of the methodology presented in this study could also be limited to the number of quality data, e.g., labeled samples. Efforts from domain experts are still needed when applying this method to a new application. However, we proposed here an approach which minimizes the labelling effort through active learning. In future work, we also aim to employ deep neural networks in both supervised and unsupervised models as more labeled samples become available in the future. The introduced model is in principle applicable to any kind of spectra, not just infrared spectra.

Author statement

Xin Tian: Methodology, Software, Formal analysis, Writing – original draft, Visualisation, Writing – review & editing. **Frederic Been:** Methodology, Conceptualization, Writing – original draft, Writing – review & editing. **Patrick Bäuerlein:** Methodology, Writing – original draft, Writing – review & editing, Conceptualization, Supervision

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2022.113569>.

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